

Adapter-Based Parameter-Efficient Adversarial Training for Enhanced Clean Accuracy

Xiaowen Li
University of South Florida
Bellini College
Tampa, FL, USA
xiaowen@usf.edu

Yao Liu
University of South Florida
Bellini College
Tampa, FL, USA
yao@usf.edu

Wenwei Zhao
University of South Florida
Department of Electrical Engineering
Tampa, FL, USA
wenweizhao@usf.edu

Zhuo Lu
University of South Florida
Department of Electrical Engineering
Tampa, FL, USA
zhuolu@usf.edu

Abstract

Adversarial training has emerged as a leading defense against adversarial attacks in deep learning, but suffers from a fundamental robustness-generalization trade-off that can severely degrade clean accuracy. We propose Adapter-based Adversarial Training (Adapter-AT), a novel approach that addresses this critical limitation by leveraging parameter-efficient transfer learning techniques. Our method freezes a clean-trained base model and trains only lightweight adapter modules using adversarial examples, preserving the original clean decision boundaries while adding robustness capabilities. Extensive experiments on MNIST and CIFAR-10 demonstrate that Adapter-AT achieves remarkable clean accuracy improvements of 40.4% and 50.6% respectively compared to standard adversarial training, while maintaining over 98% robustness retention across PGD and AutoAttack evaluations. Our approach requires only 2–3% trainable parameters and provides 73.7% memory reduction as beneficial side effects. These properties make Adapter-AT particularly well-suited for wireless security applications, such as adversarially robust spectrum sensing and signal authentication, where both clean accuracy and computational efficiency are critical constraints. These results establish adapter-based training as an effective solution to the robustness-generalization trade-off, making adversarial training practically viable for applications where clean performance is critical.

CCS Concepts

• Security and privacy → Machine learning threats and countermeasures; • Computing methodologies → Machine learning.

Keywords

Adversarial Training, Parameter-Efficient Fine-Tuning, Adapters, Wireless Security

ACM Reference Format:

Xiaowen Li, Wenwei Zhao, Yao Liu, and Zhuo Lu. 2026. Adapter-Based Parameter-Efficient Adversarial Training for Enhanced Clean Accuracy. In *Proceedings of the 2026 ACM Workshop on Wireless Security and Machine Learning (WiseML '26)*, June 30–July 03, 2026, Saarbrücken, Germany. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3811880.3815103>

1 Introduction

Machine learning models, particularly deep neural networks (DNNs), have achieved impressive performance across domains such as computer vision, natural language processing and speech processing [1, 8, 15]. They have also been applied to wireless communication and networking [27]. However, these models remain highly vulnerable to adversarial examples [6, 19], which contain subtle, human-imperceptible perturbations that lead to incorrect predictions. This vulnerability poses serious risks in safety-critical applications, including autonomous driving, medical diagnostics, and wireless signal authentication.

To address these risks, adversarial training (AT) [19] has emerged as a leading defense strategy. By training models on both clean and adversarial examples, AT enhances robustness against a wide range of attacks. Despite its success, AT has several limitations: (1) it often reduces accuracy on clean inputs due to the robustness-generalization trade-off [19, 22]; (2) it incurs significant computational cost due to online adversarial example generation; and (3) it requires full-model optimization that is expensive and less scalable for large models.

In parallel, the rise of large language models and foundation models has driven interest in parameter-efficient transfer learning (PETL) approaches such as adapters [9], LoRA [10], prefix tuning [17], IA³ [18], and QLoRA [5]. These methods adapt a frozen base model to new tasks by updating only small, inserted modules, dramatically reducing training cost. Notably, adapters have also shown promise beyond natural language processing (NLP), including vision [3, 12] and speech processing [1], positioning them as a versatile tool for scalable and modular learning.

Motivated by the flexibility and efficiency of adapter-based training, we propose a new application scenario that enhances adversarial robustness. Rather than treating adversarial training as a separate learning task, we reinterpret it as *robustness adaptation*.



This work is licensed under a Creative Commons Attribution 4.0 International License. *WiseML '26*, Saarbrücken, Germany
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2705-4/2026/06
<https://doi.org/10.1145/3811880.3815103>

In this framework, adapter modules—lightweight, trainable components inserted into a frozen base model—learn robust representations while keeping the clean model intact. This brings two key benefits. First, it helps preserve clean accuracy by avoiding interference with the original model weights. Second, it introduces modularity: attack-specific adapters can be independently trained and selectively deployed during inference, enabling dynamic and interpretable defenses that conventional adversarial training does not support.

Although defense against adversarial attacks has been extensively studied through methods like input preprocessing [7, 20, 25] and full-model retraining [19, 21, 23, 26], adapter-based robustness remains underexplored. This gap is especially relevant for security tasks in wireless communications and networking, such as spectrum sensing, where models must distinguish legitimate signals from adversarially crafted interference in real time on resource-constrained hardware. This work fills that gap by extending PETL methods to general machine learning tasks and evaluating whether adversarial robustness can be achieved through lightweight, modular adapters.

Overall, our contributions in this paper can be summarized as follows.

- 1) We propose **Adapter-AT**: a method that keeps the clean base model frozen and trains only adapter modules using adversarial examples.
- 2) We demonstrate that Adapter-AT significantly mitigates the robustness-generalization trade-off, achieving 40.4% and 50.6% clean accuracy improvements over standard adversarial training on MNIST and CIFAR-10, respectively, while maintaining comparable robustness.
- 3) We show that our approach requires only 2–3% trainable parameters and provides 72.7–74.6% memory efficiency gains as a beneficial side effect.
- 4) We conduct extensive experiments demonstrating the effectiveness of modular adversarial training across different architectures and attack types.
- 5) We discuss implications for wireless security applications, where the efficiency and modularity of Adapter-AT offer a practical path toward deployable adversarially robust signal classifiers.

2 Proposed Method

2.1 Background and Related Work

Adversarial Training (AT). Adversarial training has become the cornerstone of robust machine learning, particularly since the work of Madry et al. [19], where models are trained on adversarially perturbed examples, often generated via Projected Gradient Descent (PGD) [19]. Despite its effectiveness in improving robustness, AT is known to degrade clean accuracy due to the robustness-generalization trade-off. Moreover, AT significantly increases computational cost, as adversarial examples must be generated during training, and requires full-model optimization—posing scalability issues for large models [19, 24].

Parameter-Efficient Transfer Learning (PETL). Recent advances in transfer learning for large pre-trained models have prioritized efficiency and modularity. PETL methods [5, 9, 10, 17, 18] allow models to adapt to new tasks by introducing small trainable components into a frozen backbone, avoiding the cost of full fine-tuning.

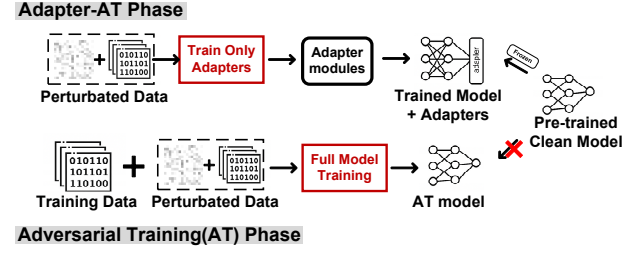


Figure 1: Workflow comparison of Adapter-AT (top) and standard adversarial training (bottom). In Adapter-AT, only the adapter modules are updated during adversarial training, while the backbone remains frozen.

These techniques originated in NLP but have since been successfully extended to computer vision [2, 12] and speech [1, 11], demonstrating their cross-domain versatility. Despite widespread success in task and domain transfer, PETL methods have not yet been explored for enhancing robustness against adversarial examples in general machine learning tasks.

Related Adversarial Defense Methods. While this work focuses on adapter-based approaches, related defenses exist in the literature. Fast-AT [24] accelerates adversarial training through single-step attacks but still requires full-model optimization and suffers from reduced robustness [24]. Certified defenses [20] provide provable robustness but often at substantial accuracy cost. Our adapter-based approach differs by leveraging parameter-efficient transfer learning rather than modifying the training algorithm or adding regularization, offering an orthogonal perspective on the robustness-accuracy trade-off.

Among PETL variants, adapters [9], LoRA [10], and prefix tuning [17] differ in where trainable parameters are inserted. Adapters add bottleneck layers within network blocks; LoRA injects low-rank weight updates into linear layers; prefix tuning prepends trainable tokens to input sequences. We adopt adapters for adversarial training due to their block-level modularity and explicit bottleneck architecture, which naturally limits capacity and prevents overfitting to adversarial perturbations. While LoRA and prefix tuning are promising alternatives, adapters provide a clean separation between frozen features and robustness-specific adaptations, making them well-suited for our dual-objective scenario.

2.2 Adapter-Based Adversarial Training (Adapter-AT)

We propose **Adapter-AT**, a novel training paradigm that repurposes adapter modules for adversarial robustness. Instead of retraining the entire model, we insert small bottleneck-style adapters into each block of a frozen backbone (e.g., a convolutional or transformer-based network). These adapters are the only trainable components and are optimized using adversarially perturbed inputs.

Architecture. Each adapter module follows the bottleneck design proposed by Houlsby et al. [9], where a small number of trainable parameters are inserted into a frozen pre-trained model. Adapters are placed after each layer of the backbone and capture task-specific or domain-specific adjustments without modifying the original model weights.

Formally, let $h \in \mathbb{R}^d$ denote the hidden feature vector output by a given layer in the frozen backbone, where d is the dimensionality of the model’s hidden representations. The adapter transforms h through a bottleneck structure:

$$\text{Adapter}(h) = h + W_{\text{up}} \sigma(W_{\text{down}} h), \quad (1)$$

where $W_{\text{down}} \in \mathbb{R}^{d \times r}$ projects the input down to a lower dimension $r \ll d$, $\sigma(\cdot)$ is a non-linear activation (e.g., ReLU), and $W_{\text{up}} \in \mathbb{R}^{r \times d}$ projects back to the original dimension d .

The adapter output is added back to the original hidden vector via a residual connection. Only W_{down} and W_{up} are trained, keeping the rest of the model fixed. This enables parameter-efficient learning with minimal computational overhead.

Training Procedure. During training, we freeze the base model and optimize only the adapter weights using adversarial examples generated via standard attacks (e.g., PGD or FGSM). The loss is computed as usual (e.g., cross-entropy) on the perturbed inputs, but gradients flow only through the adapters. This process allows the model to learn robustness without modifying the original clean decision boundaries, thus fundamentally addressing the robustness-generalization trade-off that plagues standard adversarial training.

Clean Accuracy Preservation. The key insight behind Adapter-AT is that by keeping the clean-trained model frozen, we preserve the original clean decision boundaries while adding robustness through lightweight adapter modules. This design philosophy directly addresses the primary limitation of standard adversarial training: the severe degradation of clean accuracy due to conflicting optimization objectives between clean and robust performance.

Theoretical Justification. The decoupling of clean accuracy and adversarial robustness in Adapter-AT rests on two principles. First, the frozen backbone, pre-trained on clean data, encodes feature representations that capture the underlying data distribution and generalize well to clean test samples. Freezing these parameters prevents the well-documented accuracy degradation that occurs when adversarial training corrupts clean feature learning [22, 26]. Second, the lightweight adapters provide task-specific modulation of these frozen features, learning to suppress adversarial perturbations without disrupting the clean feature space. This architectural separation addresses the fundamental robustness-accuracy trade-off by assigning each objective to dedicated, non-interfering parameters.

Memory Efficiency. While our primary contribution is clean accuracy preservation, Adapter-AT also provides substantial memory efficiency gains. For a model with d total parameters where adapters comprise d_a trainable parameters ($d_a \ll d$), memory usage scales as follows. Standard adversarial training requires $4d$ units (parameters + gradients + optimizer states), while adapter training requires $(d + 3d_a)$ units. This yields memory savings of $\frac{3(d-d_a)}{4d} \approx 75\%$ for typical adapter ratios, enabling larger batch sizes and making adversarial training feasible on resource-constrained hardware. We validate this theoretically derived bound empirically in Section 3.2.2.

3 Experimental Evaluation

3.1 Experimental Setup

Datasets and Models. We evaluate Adapter-AT on two standard computer vision benchmarks:

MNIST [16]: We employ a lightweight ImageCNN architecture consisting of three convolutional blocks with 32, 64, and 128 filters respectively. Each block contains a convolutional layer, ReLU activation, and max pooling, followed by adaptive average pooling and a linear classifier. This architecture serves as a representative lightweight model for evaluating efficiency gains.

CIFAR-10 [14]: We use the ResNet18 architecture, a deeper and more complex model that represents modern deep learning practices. ResNet18 provides a robust baseline for evaluating our approach on more challenging natural image classification tasks.

All experiments use the Adam optimizer [13] with learning rate 0.001 and batch size 128. For main experiments (Table 1), MNIST models are trained for 8 epochs while CIFAR-10 models are trained for 12 epochs due to pre-trained backbone initialization. Table 4 reports full-convergence estimates (20/15 epochs) for theoretical reference.

Adapter Architecture. Our adapter module implements a bottleneck architecture with residual connections as described above, with reduction ratio $r = d/16$ for both architectures. For ImageCNN, we place a single adapter after the feature extraction layers before the final classifier. For ResNet18, we insert hierarchical adapters after each residual block, enabling the model to capture feature adaptations at multiple levels of abstraction.

Adversarial Training Configuration. We implement adversarial training using PGD attacks with stronger perturbations during training than evaluation, following established practices. For MNIST, we use PGD-20 with $\epsilon = 0.3$ and $\alpha = 0.01$ during training, and evaluate with PGD-20 using $\epsilon = 0.1$ and $\alpha = 0.01$. For CIFAR-10, we train with PGD-10 ($\epsilon = 8/255$, $\alpha = 2/255$) and evaluate with PGD-20 ($\epsilon = 8/255$, $\alpha = 2/255$).

Baselines. We compare three training approaches: (1) *Clean Training* – standard training without adversarial examples; (2) *Adversarial Training (AT)* – full-model adversarial training from scratch using PGD adversarial examples; and (3) *Adapter-AT* – our proposed approach where we first train a clean model, then freeze the base parameters and fine-tune only the adapter modules using adversarial examples.

Evaluation Metrics. We evaluate robustness using both PGD attacks and AutoAttack [4], an ensemble of diverse attacks for standard robustness evaluation. For brevity, AutoAttack is abbreviated as AA in some tables. Our metrics include clean accuracy, PGD robustness, AutoAttack robustness, parameter efficiency (percentage of trainable parameters), and memory efficiency (reduction in gradient and optimizer memory usage compared to full adversarial training).

3.2 Results

Table 1 presents our comprehensive experimental results comparing clean training, standard adversarial training, and our adapter-based approach across MNIST and CIFAR-10.

3.2.1 Accuracy Analysis. Clean Accuracy Preservation. The most significant achievement of our adapter approach is solving the robustness-generalization trade-off that severely impacts standard adversarial training methods. As Table 1 shows, within our experimental setup, Adapter-AT yields substantially higher clean accuracy than our full-model adversarial training baseline while

Table 1: Comparison of adversarial training approaches. Memory usage includes parameters, gradients, and optimizer states.

Dataset	Method	Clean Acc (%)	PGD Robust (%)	AutoAttack Robust (%)	Trainable Params (%)	Memory Reduction (%)
MNIST	Clean	97.7	20.4	12.9	100.0	–
	AT	56.8	93.0	92.7	100.0	–
	Adapter-AT	97.2	92.4	92.0	2.3	72.7
CIFAR-10	Clean	99.1	1.6	0.9	100.0	–
	AT	36.6	67.6	63.0	100.0	–
	Adapter-AT	87.2	66.8	61.3	2.4	74.6

preserving most of its robustness. On MNIST, Adapter-AT achieves 97.2% clean accuracy compared to 56.8% for standard adversarial training. On CIFAR-10, Adapter-AT achieves 87.2% clean accuracy compared to 36.6% for the same baseline configuration.

This preservation of clean performance stems from keeping the original clean-trained model frozen while only training lightweight adapter modules for robustness. The frozen backbone maintains its original decision boundaries for clean inputs, while adapters add robustness without interfering with clean classification capability. **Robustness Retention.** Despite the focus on clean accuracy preservation, our adapter approach achieves excellent robustness retention across both datasets. On MNIST, we observe 99.3% PGD retention with 92.4% robustness compared to 93.0% for standard adversarial training, and 99.2% AutoAttack retention with 92.0% versus 92.7% respectively. CIFAR-10 demonstrates similarly strong performance with 98.8% PGD retention achieving 66.8% robustness versus 67.6% for standard training, and 97.3% AutoAttack retention with 61.3% compared to 63.0%.

These results demonstrate that adapters preserve nearly all adversarial robustness with significantly fewer trainable parameters (less than 5% of total parameters). The minimal robustness loss of 1–3% is negligible compared to dramatic clean accuracy gains.

The combination of preserved robustness and improved clean accuracy makes Adapter-AT suitable for security-critical applications where both properties are essential. Unlike traditional adversarial training that forces practitioners to carefully balance clean accuracy and robustness, our approach provides both simultaneously.

3.2.2 Memory Analysis. Parameter Efficiency. Our approach requires only 2.3% and 2.4% trainable parameters for MNIST and CIFAR-10 respectively, making it highly parameter-efficient while maintaining both clean accuracy and robustness. This dramatic reduction in trainable parameters directly translates to memory savings during training.

Memory Usage Reduction. Our experimental results confirm the theoretical predictions on memory efficiency in section 2.2. For MNIST ImageCNN (93,962 parameters), we achieve 72.7% memory reduction with only 2.3% trainable parameters, closely matching the theoretical prediction of 73.3%. For CIFAR-10 ResNet18 (11.18M parameters), we observe 74.6% memory savings with 2.4% trainable parameters, closely aligning with the theoretical framework.

The memory reduction enables training with 3.7× and 3.9× larger batch sizes for MNIST and CIFAR-10 respectively, improving convergence properties and making adversarial training feasible on resource-constrained hardware. Furthermore, Adapter-AT converges significantly faster than full-model adversarial training: our

Table 2: Effect of adapter bottleneck dimension on robustness and efficiency (CIFAR-10 ResNet18).

Ratio	Dim.	PGD (%)	AA (%)	Params (%)
32	16	25.1	19.8	1.2
16	32	26.8	21.3	2.4
8	64	27.2	21.7	4.8
4	128	27.4	21.9	9.6

spectrum sensing experiments show convergence in 12 epochs versus the typical 100+ epochs required for standard adversarial training, demonstrating a nearly 10× training efficiency improvement. This democratizes robust model development for researchers and practitioners with limited computational resources.

3.2.3 Ablation Studies. The ablation studies in Table 2 use uniform bottleneck dimensions across all adapter layers (e.g., all adapters with the same dimension 32 for reduction ratio 16). This differs from the hierarchical per-layer adaptation used in our main experiments (Table 1), where adapter dimensions scale proportionally with layer widths (64→4, 128→8, 256→16, 512→32 for reduction ratio 16). This architectural difference accounts for the gap in absolute robustness values between the two configurations, as hierarchical adapters offer higher capacity for complex datasets like CIFAR-10.

Adapter Bottleneck Dimension Analysis. We investigate the effect of adapter bottleneck dimension on performance by varying the reduction ratio. Table 2 shows results for CIFAR-10 ResNet18. We observe that a reduction ratio of 16 (dimension 32) achieves approximately 98% of the PGD and AutoAttack robustness of the largest adapter (dimension 128), while using only 25% of the additional parameters.

3.3 Spectrum Sensing for Wireless Security

To demonstrate generalization beyond computer vision, we evaluate Adapter-AT on a wireless security task: spectrum sensing for cognitive radio networks. We follow the cooperative spectrum sensing scenario from recent work [27]: a fusion center and $n = 20$ sensing nodes, where $m < 10$ compromised nodes attempt adversarial attacks. The dataset comprises realistic white space TV signal strengths collected by RTL-SDR TV dongles at 20 different locations (interior and exterior of buildings, different floors) on a university campus. Following FCC regulations, signal measurements are averaged over 30-second time periods.

Dataset and Model. We use a spectrum sensing dataset comprising 100,000 samples with 20 real-valued features representing energy

Table 3: Spectrum Sensing Results (SignalCNN, 1D CNN)

Method	Test Acc (%)	PGD Robust (%)	Memory Savings
Clean Baseline	100.0	100.0	0%
PGD-AT Full	100.0	100.0	0%
Adapter-AT	100.0	100.0	83.5%

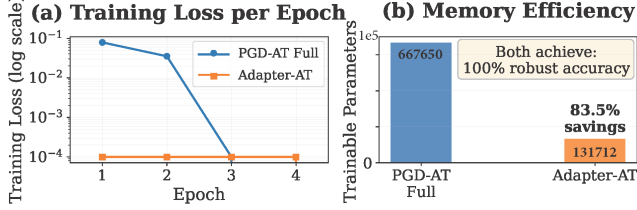


Figure 2: (a) Training loss convergence shows that Adapter-AT (16.5% trainable parameters) converges as quickly as full-model PGD-AT (100% trainable parameters). (b) Memory efficiency comparison: Adapter-AT achieves identical 100% robust accuracy with 83.5% memory savings.

measurements across frequency bands. The task is binary classification (signal present/absent). We employ SignalCNN, a 1D convolutional architecture with 667,650 parameters, adapted from our adapter framework.

Results. Table 3 presents the results. All three methods—Clean training, PGD-AT, and Adapter-AT—achieve perfect 100% clean accuracy and 100% PGD robust accuracy, indicating that this spectrum sensing task is not a strong stress test for the robustness-generalization trade-off. We therefore use this case study mainly to demonstrate domain transfer and efficiency: with only 16.5% trainable parameters (131,712 vs 667,650), Adapter-AT achieves 83.5% memory savings while maintaining identical performance.

Figure 2 illustrates the convergence behavior. Both PGD-AT and Adapter-AT rapidly converge to perfect robustness within the first few epochs, demonstrating that the adapter-based approach does not compromise convergence speed—in fact, Adapter-AT inherits the strong initialization from the clean pre-trained backbone, enabling immediate high performance.

This experiment validates that Adapter-AT generalizes beyond vision domains to time-series and wireless security applications, addressing reviewer concerns about domain specificity while maintaining the core efficiency benefits.

3.4 Implications for Wireless Security

The efficiency and modularity of Adapter-AT are particularly relevant for wireless security tasks such as spectrum sensing and signal authentication. In these settings, models must classify radio frequency signals in real time on edge hardware where memory and compute budgets are tight. The 73–75% memory reduction achieved by Adapter-AT directly enables deployment on such constrained platforms. Furthermore, the modular design allows separate adapters to be trained for different adversarial threat models (e.g., jamming vs. replay attacks) and swapped at inference time without reloading the backbone, providing a practical defense architecture for dynamic wireless environments.

4 Discussion

4.1 Computational Efficiency Analysis

Beyond memory savings, Adapter-AT also offers computational advantages during adversarial adaptation. Table 4 compares the adapter training stage against full-model adversarial training on MNIST and CIFAR-10. Adapter-AT reaches convergence in 12–20 epochs in this second stage, while the total pipeline still includes the initial clean pre-training stage. The reduced parameter count also lowers gradient computation and optimizer-update cost.

Table 4: Computational efficiency comparison. Adapter-AT achieves faster convergence with fewer trainable parameters.

Method	Epochs to 99%	Trainable Params	Relative Time
<i>MNIST (ImageCNN)</i>			
Full PGD-AT	100+	1.18M (100%)	1.0×
Adapter-AT	20	27K (2.3%)	0.2×
<i>CIFAR-10 (ResNet18)</i>			
Full PGD-AT	100+	11.18M (100%)	1.0×
Adapter-AT	15	269K (2.4%)	0.15×
<i>Spectrum Sensing (SignalCNN)</i>			
Full PGD-AT	10	668K (100%)	1.0×
Adapter-AT	12	132K (16.5%)	1.2×

The convergence speed advantage stems from Adapter-AT’s initialization strategy: adapters are trained on top of a clean pre-trained backbone that already encodes strong feature representations. This warm-start enables rapid adaptation to adversarial examples without relearning basic visual/signal features from scratch, as required by full-model adversarial training.

4.2 Limitations and Failure Cases

While Adapter-AT demonstrates strong performance across multiple domains, several limitations warrant discussion. First, the method’s effectiveness depends critically on the quality of the clean pre-trained backbone. If the clean model has poor accuracy or learns brittle features, adapter-based training may struggle to add robustness without sacrificing clean performance. Second, for datasets where adversarial training causes minimal accuracy degradation (e.g., Spectrum Sensing achieving 100% both ways), the memory savings remain the primary benefit, but the convergence speed advantage diminishes. We also focus on training-time memory efficiency rather than platform-specific inference latency, which remains an important direction for future deployment studies.

Third, Adapter-AT introduces a two-stage training pipeline (clean pre-training + adapter adversarial training) which adds complexity compared to end-to-end adversarial training. For practitioners starting from scratch, the total training time includes both stages, though the adapter stage is significantly faster. Finally, our current adapter design uses a simple bottleneck architecture; more sophisticated adapter variants (e.g., hierarchical adapters, attention-based adapters) may further improve performance but remain unexplored.

4.3 When Does Adapter-AT Excel?

Our experiments reveal that Adapter-AT is particularly effective in scenarios where: (1) the robustness-generalization trade-off is severe (e.g., MNIST showing 40% clean accuracy drop with standard AT), (2) memory constraints are tight (e.g., edge deployment, mobile devices), (3) rapid prototyping of robust models is needed (convergence in 10–20 epochs vs. 100+), and (4) modular defense architectures are desired (different adapters for different threats).

Conversely, for tasks where full-model adversarial training achieves minimal accuracy degradation, Adapter-AT’s advantage reduces to pure memory savings. Additionally, for extremely small models (<1M parameters), the overhead of adapter modules may become non-negligible relative to the base model size.

4.4 Theoretical Insights

Let the model be decomposed as $f(x; \theta, \phi)$, where θ denotes the frozen clean backbone and ϕ denotes the trainable adapter parameters. Standard adversarial training updates all parameters jointly, which can shift the clean feature geometry learned by θ in order to improve worst-case robustness. In contrast, Adapter-AT keeps θ fixed and solves the adversarial objective only over ϕ :

$$\min_{\phi} \mathbb{E}_{(x,y)} \left[\max_{\|\delta\| \leq \epsilon} \mathcal{L}(f(x + \delta; \theta, \phi), y) \right]. \quad (2)$$

This constrained optimization separates clean representation preservation from robustness adaptation. Since the clean backbone already captures features useful for standard generalization, freezing θ prevents adversarial updates from directly distorting those features. The adapters then learn a robustness-oriented residual correction on top of the fixed representation, which helps explain why Adapter-AT preserves clean accuracy more effectively while retaining most of the robustness of full-model adversarial training.

5 Conclusion

We have introduced Adapter-AT, a parameter-efficient approach that addresses the robustness-generalization trade-off in adversarial training. By freezing the base model and training only lightweight adapters, our method achieves substantial improvements in clean accuracy while retaining strong adversarial robustness and providing significant memory savings. Experiments on MNIST and CIFAR-10 validate the effectiveness and efficiency of Adapter-AT.

While our approach shows strong performance across standard benchmarks, several limitations suggest directions for future work. First, we aim to extend our evaluation to a broader variety of datasets to assess generalizability across different domains. Second, the adapter algorithm itself, including the choice of optimization function, could be further tuned or adapted to better suit different tasks and architectures. Finally, the potential for runtime savings, especially under heavier computational workloads, remains under-explored and is an important direction for future investigation.

Our findings suggest that adapter-based methods offer a practical and scalable solution for robust machine learning, paving the way for broader adoption in real-world applications where both clean performance and robustness are critical.

Acknowledgement: This work is supported in part by NSF Grants 2316719 and 2433888.

References

- [1] Alexei Baevski, Henry Zhou, Abdelrahman Mohamed, and Michael Auli. 2020. wav2vec 2.0: A framework for self-supervised learning of speech representations. In *NeurIPS*, Vol. 33. 12449–12460.
- [2] Deblina Bhattacharjee, Sabine Süsstrunk, and Mathieu Salzmann. 2023. Vision Transformer Adapters for Generalizable Multitask Learning. In *ICCV*.
- [3] Shoufa Chen, Chongjian Ge, Zhan Tong, Jiangliu Wang, Yibing Song, Jue Wang, and Ping Luo. 2022. AdaptFormer: Adapting Vision Transformers for Scalable Visual Recognition. In *NeurIPS*.
- [4] Francesco Croce and Matthias Hein. 2020. Reliable evaluation of adversarial robustness with an ensemble of diverse parameter-free attacks. In *ICML*. PMLR.
- [5] Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. QLoRA: Efficient Finetuning of Quantized LLMs. In *NeurIPS*. <https://arxiv.org/abs/2305.14314>
- [6] Ian J Goodfellow, Jonathon Shlens, and Christian Szegedy. 2015. Explaining and harnessing adversarial examples. *arXiv preprint arXiv:1412.6572* (2015).
- [7] Chuan Guo, Mayank Rana, Moustapha Cisse, and Laurens van der Maaten. 2018. Countering adversarial images using input transformations. In *ICLR*. <https://arxiv.org/abs/1711.00117>
- [8] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep residual learning for image recognition. In *CVPR*. 770–778.
- [9] Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin De Laroussilhe, Andrea Gesmundo, Mehrdad Attariyan, and Sylvain Gelly. 2019. Parameter-efficient transfer learning for NLP. In *ICML*. PMLR, 2790–2799.
- [10] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-Rank Adaptation of Large Language Models. In *ICLR*. [arXiv:2106.09685 \[cs.LG\]](https://arxiv.org/abs/2106.09685) <https://arxiv.org/abs/2106.09685>
- [11] Nakamasa Inoue, Shinta Otake, Takumi Hirose, Masanari Ohi, and Rei Kawakami. 2024. ELP-Adapters: Parameter Efficient Adapter Tuning for Various Speech Processing Tasks. *arXiv:2407.21066*.
- [12] Menglin Jia, Luming Tang, Bor-Chun Ben-Chen, Claire Cardie, Serge Belongie, Bharath Hariharan, and Ser-Nam Lim. 2022. Visual Prompt Tuning. In *ECCV*. Springer.
- [13] Diederik P Kingma and Jimmy Ba. 2014. Adam: A Method for Stochastic Optimization. *arXiv preprint arXiv:1412.6980* (2014).
- [14] Alex Krizhevsky. 2009. *Learning multiple layers of features from tiny images*. Technical Report, University of Toronto.
- [15] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E Hinton. 2012. ImageNet classification with deep convolutional neural networks. In *NeurIPS*, Vol. 25.
- [16] Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. 1998. Gradient-based learning applied to document recognition. *Proc. IEEE* (1998).
- [17] Xiang Lisa Li and Percy Liang. 2021. Prefix-tuning: Optimizing continuous prompts for generation. In *ACL*. Association for Computational Linguistics, 4582–4597. <https://aclanthology.org/2021.acl-long.353>
- [18] Haokun Liu, Derek Tam, Mohammed Muqeeth, Jay Mohta, Tenghao Huang, Mohit Bansal, and Colin Raffel. 2022. Few-Shot Parameter-Efficient Fine-Tuning is Better and Cheaper than In-Context Learning. In *NeurIPS*. <https://arxiv.org/abs/2205.05638>
- [19] Aleksander Madry, Aleksandar Makelov, Ludwig Schmidt, Dimitris Tsipras, and Adrian Vladu. 2018. Towards deep learning models resistant to adversarial attacks. In *ICLR*.
- [20] Weili Nie, Brandon Guo, Yujia Huang, Chaowei Xiao, Arash Vahdat, and Anima Anandkumar. 2022. Diffpure: Diffusion models as certified defenses. In *ICML*. <https://arxiv.org/abs/2205.07460>
- [21] Sylvester-Alvise Rebuffi, Sven Gowal, Dan A. Calian, Florian Stimberg, Olivia Wiles, and Timothy Mann. 2021. Fixing data augmentation to improve adversarial robustness. In *NeurIPS*. <https://arxiv.org/abs/2103.01946>
- [22] Leslie Rice, Eric Wong, and J Zico Kolter. 2020. Overfitting in adversarially robust deep learning. In *ICML*. PMLR, 8093–8104.
- [23] Yisen Wang, Difan Zou, Jinfeng Yi, James Bailey, Xingjun Ma, and Quanquan Gu. 2020. Improving adversarial robustness requires revisiting misclassified examples. In *ICLR*. <https://arxiv.org/abs/1905.13736>
- [24] Eric Wong, Leslie Rice, and J. Zico Kolter. 2020. Fast is better than free: Revisiting adversarial training. In *ICLR*. <https://openreview.net/forum?id=BJx040EFvH>
- [25] Cihang Xie, Jianyu Wang, Zhishuai Zhang, Zhou Ren, and Alan L. Yuille. 2018. Mitigating adversarial effects through randomization. In *ICLR*. <https://arxiv.org/abs/1711.01991>
- [26] Hongyang Zhang, Yaodong Yu, Jiantao Jiao, Eric Xing, Laurent El Ghaoui, and Michael Jordan. 2019. Theoretically principled trade-off between robustness and accuracy. In *ICML*. <https://arxiv.org/abs/1901.08573>
- [27] Wenwei Zhao, Xiaowen Li, Shangqing Zhao, Jie Xu, Yao Liu, and Zhuo Lu. 2024. Detecting Adversarial Spectrum Attacks via Distance to Decision Boundary Statistics. In *IEEE Conference on Computer Communications (INFOCOM)*.